

BSAI – 7TH SEMESTER

PRESENTATION 4 – WRITING PROCESS

7TH OCTOBER 2025

PROPOSAL

- PROPOSAL
- A formal plan / document that outlines WHAT, WHY, & HOW of your project
- Main purpose is to persuade / convince others (supervisors / sponsors / clients etc.)
- EXAMPLES
 - 1. Enhancing Chatbot Accuracy
 - 2. AI-Powered Classroom Attendance System
 - 3. AI-based Health Monitoring App

RESEARCH PROPOSAL

- A technical document
- A planned study / investigation
- **STEPS / STAGES**
- 1. Research Problem
- 2. Objectives
- 3. Hypothesis / Research Questions
- 4. Methodology
- 5. Expected Outcomes

SAMPLE RESEARCH PROPOSALS

TOPIC	RESEARCH PROPOSAL
1.AI in Healthcare	This study aims to develop an AI model that predicts early signs of diabetes using patient data. Machine learning algorithms will analyze medical records to improve early diagnosis accuracy.
2. AI in Education	This research investigates how AI chatbots can support students' learning by providing personalized feedback and adaptive learning materials.
3. Sentiment Analysis	This project focuses on developing an NLP model that analyzes social media posts to determine public sentiment toward emerging technologies.

WRITING PROPOSALS - EXERCISE

- **TOPIC:** AI for Cybersecurity
- **TASK:** *Draft a short proposal for an AI tool that detects phishing emails.*
- **PROBLEM:** Identify the problem which needs a solution (Research gap)
- (Increasing phishing attacks make it difficult for users to distinguish between genuine and fake emails.)
- **OBJECTIVE:** What do you aim to achieve?
- (To develop an AI-based system that accurately detects phishing attempts in emails.)
- **METHODOLOGY:** How will you collect and analyze data?
- (Collect and label email datasets as “safe” or “phishing,” extract key features, and train a supervised learning model using Scikit-learn.)

TYPES OF PROPOSALS (GENERAL)

- **1. Project Proposal**
 - **Purpose:** To suggest a plan for developing a product, system, or service.
 - **Example:** *Designing an AI-based attendance system using facial recognition.*
- **2. Business Proposal**
 - **Purpose:** To pitch an idea or product to investors or clients for funding or partnership.
 - **Example:** *Launching a mobile app that uses AI to recommend personalized study plans for students.*
- **3. Technical Proposal**
 - **Purpose:** To present a technical solution to a specific problem.
 - **Example:** *Developing an AI tool that detects network intrusions in real time.*

RESEARCH PROPOSALS (ACADEMIC)

- **1. Academic Research Proposal**
- **Purpose:** To outline a study for academic approval or a degree requirement.
- **Example:** *A proposal to analyze how deep learning improves image recognition accuracy.*
- **2. Scientific/Experimental Research Proposal**
- **Purpose:** To test hypotheses using experiments or data analysis.
- **Example:** *Investigating how AI models detect phishing emails using supervised learning.*
- **3. Analytical Research Proposal**
- **Purpose:** To examine data, theories, or existing research to identify patterns or solutions.
- **Example:** *Analyzing AI bias in facial recognition datasets.*

RESEARCH PROPOSALS TYPES – CONTD.

- **4. Descriptive Research Proposal**

- **Purpose:** To describe phenomena or characteristics without manipulating variables.
- **Example:** *Studying how AI tools are currently used in educational settings.*

- **5. Comparative Research Proposal**

- **Purpose:** To compare two or more methods, models, or systems.
- **Example:** *Comparing the performance of CNN and RNN models for speech recognition.*

- **6. Applied Research Proposal**

- **Purpose:** To solve real-world problems through practical application.
- **Example:** *Using AI to optimize energy consumption in smart homes.*

CHARACTERISTICS OF A PROPOSAL

C-PROF-FACTS

- **C – Clarity** (clear purpose and goals)
- P – Persuasiveness** (convincing and logical argument)
- R – Relevance** (focused on the main problem or need)
- O – Objectives** (specific and measurable aims)
- F – Feasibility** (practical and achievable plan)
- F – Formal tone** (professional and academic writing style)
- A – Accuracy** (technical and factual correctness)
- C – Conciseness** (brief and direct communication)
- T – Technicality** (use of correct terms and structured format)
- S – Structure** (organized and coherent layout)